

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The UDP protocol reveals that: The browser's attempt to request the IP address for the domain "www.yummyrecipesforme.com" is failing because the packets are not being received by the DNS service.

This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message: "udp port 53 unreachable".

The port noted in the error message is used for: DNS (Domain Name System) service.

The most likely issue is: The DNS server is offline, or the DNS service is not actively listening on port 53 to resolve domain names into IP addresses.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

Time incident occurred: 1:24 p.m. (13:24:32.192571)

Explain how the IT team became aware of the incident: Several customers reported that they were unable to access the client company's website (www.yummyrecipesforme.com) and received a "destination port unreachable" error after waiting for the page to load.

Explain the actions taken by the IT department to investigate the incident: A cybersecurity analyst attempted to visit the website to replicate the error. To troubleshoot, the analyst loaded the network analyzer tool, tcpdump, and attempted to load the webpage again to capture and inspect the network traffic packets.

Note key findings of the IT department's investigation (i.e., details related to the port affected, DNS server, etc.): The tcpdump logs revealed that the initial outgoing UDP request from the client's computer to the DNS server (to retrieve the IP address for the website) failed. In response to the UDP packet, the DNS server (IP 203.0.113.2) returned an ICMP packet containing the error message "udp port 53 unreachable." Since port 53 is the standard port for DNS services, the "unreachable" message indicates that no service was listening on the receiving end.

Note a likely cause of the incident: A likely cause is that the DNS service on the server has crashed or is stopped. Alternatively, a misconfigured firewall could be blocking incoming UDP traffic on port 53 at the DNS server level.